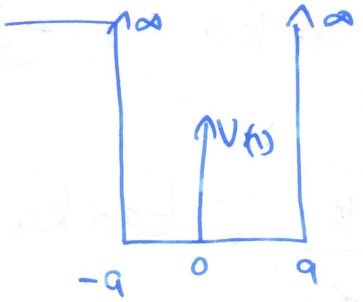
a) $V(x) = \alpha \delta(x)$



$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

For bound states, with $E<0$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi(x)$$

$$\frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2}\psi = 0 \Rightarrow \frac{d^2\psi}{dx^2} + k^2\psi = 0$$

$$k = \sqrt{\frac{2mE}{\hbar^2}}.$$

The general solution can be written as

$$\psi(x) = A\cos kx + B\sin kx \quad \text{or}$$
$$A'\cos k(x-a) + B\sin k(x-a) \Bigg\} \; x>0$$

$\psi(x) = 0$ at $x = \pm a$

$\Rightarrow A' = 0$

$$\psi(x) = \begin{cases} B\sin k(x-a) & x>0 \\ -B\sin k(x+a) & x<0 \end{cases}$$

At $x=0$, $\psi(0) = -B\sin ka$

$$-\frac{\hbar^2}{2m}\int_{-\epsilon}^{\epsilon}\frac{d^2\psi}{dx^2}dx + \int_{-\epsilon}^{\epsilon}V(x)\psi(x)\,dx = \int_{-\epsilon}^{\epsilon}E\psi(x)\,dx$$

$$\Rightarrow \left(\frac{d\psi}{dx}\right)\Bigg|_{0^-}^{0^+} = \frac{2m\alpha\psi(0)}{\hbar^2}$$

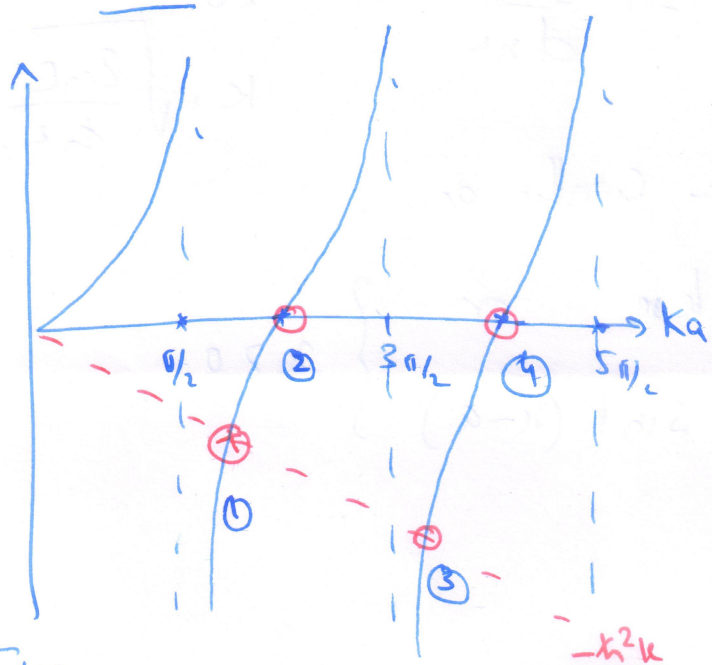
$$2Bk\cos ka = +\frac{2m\alpha}{\hbar^2}(-B\sin ka)$$

$$-\frac{\hbar^2 k}{m\alpha} = \tan ka \longrightarrow \text{Even solution}$$

$$0 = \sin ka \longrightarrow \text{odd solution}$$

or $(\psi(0) = 0)$ (Similar as ∞ potential well)

$\alpha > 0$



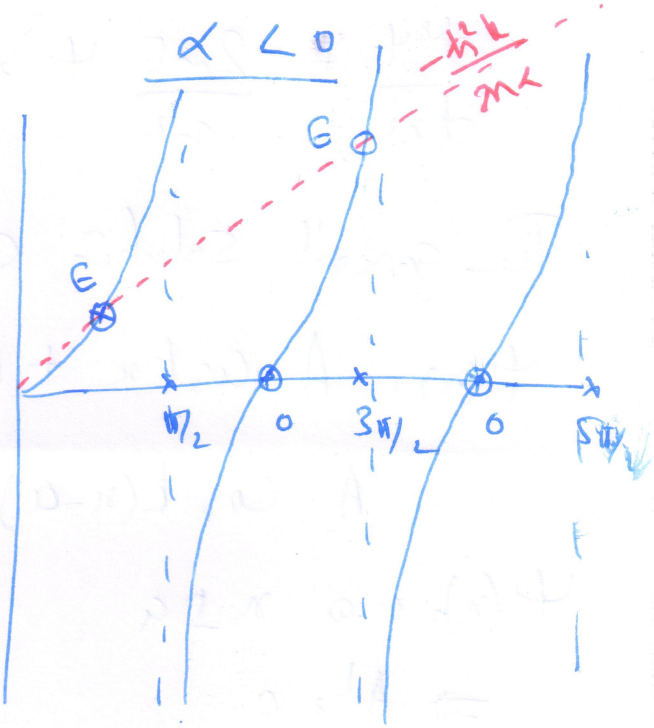
Even eigenvalues occur when $ka > (2n+1)\frac{\pi}{2}$

Odd eigenvalue occur when $ka = (2n+2)\frac{\pi}{2}$

$n = 0, 1, 2, \ldots$

Even eigenvalues are at a higher level compared to ∞ well with no $\delta(x)$ at middle.

$\alpha < 0$

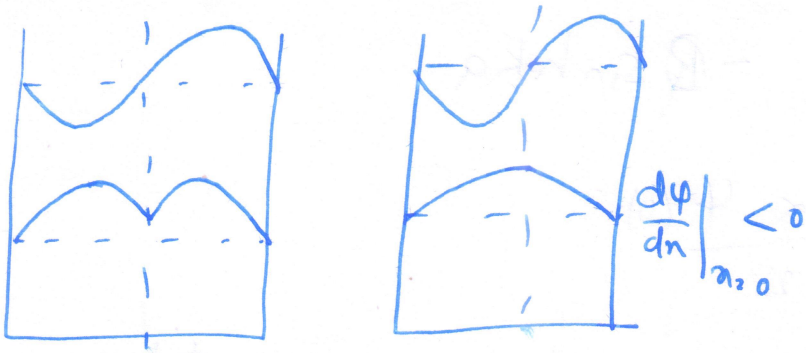


Even eigen values occur $ka < (2n+1)\pi/2$

Odd eigen values occur when $ka = (2n+2)\pi/2$

$n = 0, 1, 2 \ldots$

Even eigen values are at a lower level compared to a well with no $\delta(x)$ at middle.



$\alpha > 0$ $\qquad$ $\alpha < 0$

$\left.\frac{d\psi}{dx}\right|_{x=0} > 0.$ $\qquad$ $\left.\frac{d\psi}{dx}\right|_{x=0} < 0.$

For $\alpha < 0$, there could exist a bound state with $E < 0$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi(x)$$

$$\frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2}\psi = 0 \qquad \text{Here } E < 0$$

$\psi(x) \to 0$ as $x \to \pm a$.

The general solution can be written as $\qquad k = \sqrt{\frac{-2mE}{\hbar^2}}, \; E < 0$

$\psi(x) = Ae^{+kx} + Be^{-kx}$

or $A''\cosh kx + B''\sinh kx$

or Since $\psi(x) \to 0$ as $x \to \pm a$

or $P\cosh k(x-a) + \Phi\sinh k(x-a) \qquad x > 0$

$P\cosh k(x+a) \pm \Phi\sinh k(x+a) \qquad x < 0$

Since $\psi(x) \to 0$ as $x \to \pm a$

$P = 0$

$\psi(x) = \Phi\sinh k(x-a) \qquad x > 0$

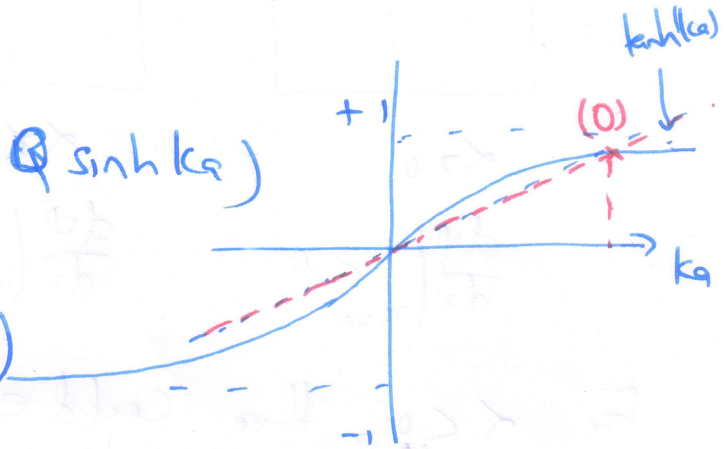
$-\Phi\sinh k(x+a) \qquad x < 0$

At $x = 0$, $\psi(0) = -\Phi \sinh ka$

$$\left.\frac{d\psi}{dx}\right|_{0^-}^{0^+} = \frac{2m\alpha\,\psi(0)}{\hbar^2}$$

$$2\Phi k \cosh ka = \frac{2m\alpha}{\hbar^2}\left(-\Phi \sinh ka\right)$$

$$\tanh ka = -\frac{\hbar^2 k}{m\alpha} \quad (\alpha < 0)$$



The intersection gives k value of bound state in delta function well

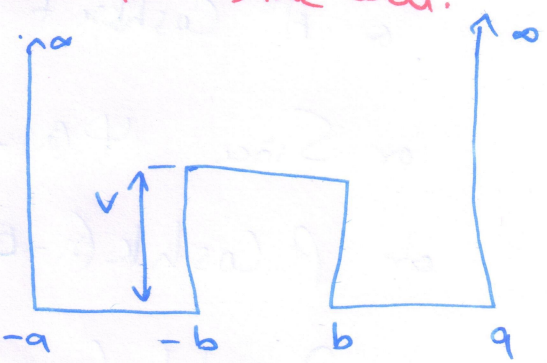
$$\psi(x) = \pm\Phi \sinh k(x-a) \quad x > 0$$
$$-\Phi \sinh k(x+a) \quad x < 0$$

Φ can be found by $\int_{-a}^{a} |\psi|^2 dx = 1$.

If α is a small -ve number, there might be an intersection and no bound state inside the Dirac well.

c)

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \psi(x)V(x) = E\psi(x)$$



$$\frac{d^2\psi}{dx^2} + k^2\psi = 0 \quad \text{[outside barrier]}$$

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\frac{d^2\psi}{dx^2} - k_1^2\psi = 0 \quad \text{[inside barrier]} \quad \text{for } 0 < E < V$$

$$k_1 = \sqrt{\frac{2m(V-E)}{\hbar^2}}$$

Using the arguments in (a) we can write the general solution as

$$\psi(x) = \begin{cases} B\sin k(x-a) & b < x < a \\ -B\sin k(x+a) & -a < x < -b \end{cases}$$

$$\psi(x) = \begin{cases} P\cosh k_1 x & -b < x < b \quad \text{Even} \\ Q\sinh k_1 x & -b < x < b \quad \text{Odd} \end{cases}$$

Since ψ & ψ' should be continuous at $x = \pm b$

$P\cosh k_1 b = B\sin k(b-a)$ Even

$Q\sinh k_1 b = B\sin k(b-a)$ Odd

$$\therefore P = \frac{B\sin k(b-a)}{\cosh k_1 b} \qquad Q = \frac{B\sin k(b-a)}{\sinh k_1 b}$$

$$\left.\frac{d\psi}{dx}\right|_{x = \pm b}$$

$Pk_1\sinh k_1 b = Bk\cos k(b-a)$ Even

$Qk_1\cosh k_1 b = Bk\cos k(b-a)$ Odd

$$k_1\frac{\sinh k_1 b \cdot B\sin k(b-a)}{\cosh k_1 b} = Bk\cos k(b-a)$$

$$\frac{B\sin k(b-a)}{\sinh k_1 b}\, k_1\cosh k_1 b = Bk\cos k(b-a)$$

$\frac{k_1}{k} \tanh k_1 b = \cot k(b-a)$ Even

$\frac{k_1}{k} \coth k_1 b = \cot k(b-a)$ Odd.

Solving these two transcendental equations given k values of odd & Even Solutions.

B can be found by $\int_{-a}^{\infty} |\psi|^2 dx = 1$.

